TOPIC 2 · PERSONALISATION, .RULES AND SKILLS FOR EXCEL
LAB 6 THEME · REPORTING SKILL

Lab 6 — Create, Use and Improve the Maintenance Reporting Skill

Maps to	LO2: distinguish account Personalisation, workbook .Rules and reusable Skills; configure, apply and verify each using water-service evidence.
Duration	30 minutes
Tools	Excel and PowerPoint Copilot · Microsoft 365 Copilot web · OneDrive
Mock data	data/Northstar_Work_Orders.xlsx
Lab files	data/templates/maintenance-reporting/SKILL.md (Reviewed reporting Skill fallback)
Course	GenAI for Engineers · v2.3
Prompts	https://tertiarycourses.github.io/IES-GenAI-for-Engineers/prompts/#lab-6

Goal

Reuse the Skill-authoring and installation method from Lab 5 to create maintenance-reporting. Apply it in Excel and PowerPoint, test missing inputs and a controlled rerun, then refine v2 using actual evidence. The supplied reviewed template supports rejoining; it is not proof of native execution.

What You Will Do

Reuse the Skill-authoring and installation method from Lab 5 to create maintenance-reporting. Apply it in Excel and PowerPoint, test missing inputs and a controlled rerun, then refine v2 using actual evidence. The supplied reviewed template supports rejoining; it is not proof of native execution.

What You Will Build

Northstar_Skill_Report.xlsx with Maintenance_Summary and Assignment_Proposals, Northstar_Skill_Update.pptx, SKILL.md v2 and a before/after improvement log.

Prerequisites

- Labs 1–2 verified brief and four-slide update; Lab 5 Skill-authoring method.

Data note

Northstar Water Services is a fictional water-pipe maintenance and service company. All service hubs, staff and work orders are synthetic. Use the fixed reporting date 2026-09-30; missing facts remain UNKNOWN. Proposals do not change real assignments. Training email drafts are never sent.

Steps

Step 1. Use Microsoft 365 Copilot on the web to draft maintenance-reporting/SKILL.md from the complete specification below. Review it against data/templates/maintenance-reporting/SKILL.md, save the complete file as v1 in work-genai, then install it using the same Excel app-created custom Skills folder and Refresh route practised in Lab5. If a native Skill menu is missing, use the full reviewed file as pasted instructions, remove @mention and label the route manual.

PROMPT / CHECK — copy and adapt

```
MICROSOFT 365 COPILOT WEB:
Write complete maintenance-reporting SKILL.md with YAML name/description. Require actual
tblWorkOrders or a verified Word snapshot, report date and filter scope. Respect visible
.Rules. Define backlog, overdue and Closed-only on-time rate. Preserve IDs/statuses/current
owners; keep assignment proposals separate and approvals UNKNOWN. Create a verified Excel
summary and improve a supplied four- or eight-slide PowerPoint report from selected evidence.
Ask for missing inputs and before replacing only named outputs; rerun without duplicates or
source changes. Keep the existing design and source/date notes. Never send email or approve
assignments. Return the entire file.
```

Step 2. Copy the Lab 6 reviewed SKILL.md to your output folder and retain v1. Open this lab's clean workbook or your Lab 1 cleaned workbook. Save as Northstar_Skill_Report.xlsx. Read Technicians: primary category and available hours are synthetic planning inputs, not approvals. Check that your skill is enabled in Excel.

Step 3. Invoke the skill in Plan or Chat only, review its proposed output boundaries, then allow editing. Inspect the resulting formulas and charts. Summary counts must reconcile to Control, and the source table must remain 120 unique records.

PROMPT / CHECK — copy and adapt

```
EXCEL - PLAN:
@maintenance-reporting Build Maintenance_Summary for tblWorkOrders at 2026-09-30 with all
filters All. Include total, closed, cancelled, backlog, overdue, on-time closed rate and
recorded cost; show formulas and source scope. Plan first. Preserve source records.

EXCEL - ALLOW EDITING:
Apply that reviewed summary plan. Add Assignment_Proposals for unassigned backlog only. Use
tblTechnicians primary category and available hours as planning inputs; show category/capacity
checks, current owner and proposed owner separately. Leave manager decision UNKNOWN. Do not
change Assigned_To.
```

Step 4. Check the unassigned backlog count against Control. A technician with the same category may still lack usable capacity: use available hours minus known proposed hours, flag conflicts, and do not convert a suggestion into confirmed assignment. Copy your verified summary into a copy of data/checkpoints/Northstar_Verified_Snapshot.docx, preserving measures and date.

PROMPT / CHECK — copy and adapt

```
EXCEL - CHAT ONLY:
@maintenance-reporting Review Assignment_Proposals against tblWorkOrders and tblTechnicians.
Check each suggested owner has the matching water-service job category, capacity limitations
are visible and original Assigned_To values are unchanged. Leave manager approval UNKNOWN; do
not assign or schedule a technician.
```

Step 5. In PowerPoint select the reviewed Word snapshot and open the four-slide update from Lab2 or data/checkpoints/Northstar_Initial_Update_Checkpoint.pptx. Use Copilot > + > Choose skills > Manage skills > Add skill to upload the reviewed maintenance-reporting file, or use the app's own supported Skills folder. Verify it in PowerPoint separately from Excel. If missing, paste the complete reviewed Skill plus snapshot as manual instructions and remove @mention. Apply the following refinement after review.

PROMPT / CHECK — copy and adapt

```
POWERPOINT COPILOT:
@maintenance-reporting Improve this existing four-slide Northstar update from the selected
verified Word snapshot. Preserve slide count, all values and source/date/scope notes; use
clear claim headlines and the existing design. Keep assignment suggestions explicitly proposed
and do not assume access to Excel chat.
```

Step 6. Create a new EMPTY workbook with no table, open a fresh Copilot chat and run a missing-input test for a different reporting period with no source selected. A proper skill asks for source/date/scope rather than inventing metrics. If it guesses, record the failure and revise the skill's input guard. Return to Northstar_Skill_Report.xlsx and Save As Northstar_Skill_Rerun_Test.xlsx. Run the second summary request in this copy of the report workbook, not in the empty workbook; only named outputs may be replaced after review, with no duplicated source or output sheets. Compare a before/after copy of Work_Orders sorted by Work_Order_ID for status, owner, dates and cost; every original source value must match.

PROMPT / CHECK — copy and adapt

```
EXCEL - CHAT ONLY:
@maintenance-reporting Prepare my next maintenance report. I need a DIFFERENT reporting period
from the course example and have not supplied any source, date or scope. This empty workbook
contains no work orders. Tell me what you need; do not invent data.

EXCEL - PLAN:
@maintenance-reporting Rebuild the existing Maintenance_Summary for the same table, date and
scope. Describe what will be replaced, ask before replacement and keep source/unrelated
sheets. Do not duplicate outputs.
```

Step 7. In a plain text editor improve SKILL.md: require an input guard, category/capacity evidence, source provenance per claim and a clear rerun rule. Retain YAML name maintenance-reporting and add a body version note v2. Re-upload/Refresh in BOTH Excel and PowerPoint. Record actual before/after changes, or already present - retained for safeguards that your v1 already contains. Do not fabricate differences. Rerun the smoke tests and compare values. If using the manual route, replace the pasted instructions in both apps.

PROMPT / CHECK — copy and adapt

```
MICROSOFT 365 COPILOT WEB:
Improve this supplied maintenance-reporting skill to v2 using only my actual smoke-test and
rerun findings pasted here. Add explicit missing-input and no-duplicate-output guards. Keep
Excel source and current owners protected. Return the complete revised SKILL.md and a short
before/after change log; do not fabricate test evidence.
```

Sample Prompts

Use the app labelled for each prompt (or adapt it). Name the table and columns, state the output, and always verify the result. Use the linked prompts web page for one-click Copy buttons.

Copy-ready prompts: https://tertiarycourses.github.io/IES-GenAI-for-Engineers/prompts/#lab-6

#	Purpose	Prompt
1	Scope	State actual source, report date and selected scope before acting.
2	Verify	Explain the source/formula checks and preserve unknowns and original records.

Test It

Check	Expected result
Source preserved	120 unique orders; current owner/status unchanged.
Skill outputs	Backlog 60; overdue 50; on-time closed 30/48; cost S$35,827.50.
Improvement evidence	Input guard and rerun tested; v1/v2 retained; PowerPoint four-slide output accurate.

Checkpoint

If you fell behind, use data/checkpoints/Northstar_Verified_Checkpoint.xlsx and data/checkpoints/Northstar_Verified_Snapshot.docx. Record checkpoint used in Prompt_Log; then perform this lab, rather than claiming the checkpoint as your output.

Troubleshooting

If this happens	Fix
Copilot cannot find the source	Use the app file picker to select the actual OneDrive file. A typed local path or a different app chat is not shared context. If needed copy only the verified snapshot into the current app.
A total disagrees with Control	Clear dashboard filters to All, check table extent and duplicate IDs, recalculate, then compare definitions and reporting date. Reconcile before continuing.
The named feature is missing	Confirm the correct assigned account, English display language and app version. Use the stated manual practice route; record feature unavailable. This is not evidence that a native skill ran.

Challenge

Repeat the task for one different site and explain how the filter scope changes the result. Keep the original verified output.

Reflection

Which evidence proves the output is correct, and which decision still needs a person?